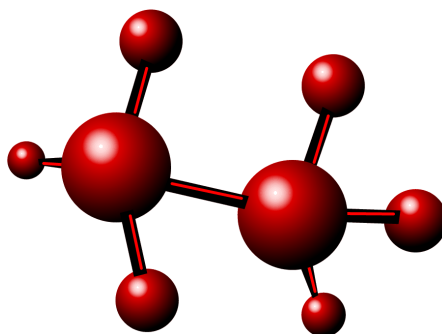




SCIENCE

Invitational A • 2016



GENERAL DIRECTIONS:

- DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
- Ninety minutes should be ample time to complete this contest, but since it is not a race, contestants may take up to two hours. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet.
- You may place as many notations as you desire anywhere on the test paper except on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- On the back of this page is printed a copy of the periodic table of the elements. You may wish to refer to this table in answering the questions, and if needed, you may use the atomic weights and atomic numbers from the table. Other scientific relationships are listed also.
- Silent hand-held calculators that do not need external wall plugs may be used. Graphing calculators that do not have built-in or stored functionality that provides additional scientific information are allowed. Small hand-held computers are not permitted. Calculators that accept memory cards or memory sticks are not permitted. Each contestant may bring one spare calculator. All memory must be cleared.

SCORING:

All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for an incorrect answer.

Periodic Table of the Elements

1A 1		2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.008																		2 He 4.003
3 Li 6.941	4 Be 9.012												5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12		13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.41		31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41		49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59		81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (272)	112 Cn (285)							

58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Some Standard Properties of Water

property	symbol	value
density of water	ρ_{water}	1.000 g cm ⁻³
density of ice	ρ_{ice}	0.9167 g cm ⁻³
specific heats		
ice	C_{ice}	2.09 J g ⁻¹ K ⁻¹
water	C_{water}	4.184 J g ⁻¹ K ⁻¹
steam	C_{steam}	2.03 J g ⁻¹ K ⁻¹
heat of fusion	ΔH_{fus} or L_f	334 J g ⁻¹
heat of vaporization	ΔH_{vap} or L_v	2260 J g ⁻¹
index of refraction	n	1.33
autoionization	K_w	1.0×10^{-14}

Pressure		
1 atm	=	760 torr
	=	101325 Pa
	=	14.7 psi
1 bar	=	10 ⁵ Pa
	=	100 kPa

Energy		
1 cal	=	4.184 J
1 L atm	=	101.325 J
1 Cal	=	1 kcal
1 hp	=	746 W
1 eV	=	1.602×10^{-19} J

Various Physical Constants

property	symbol	value
universal gas constant	R	8.314 J mol ⁻¹ K ⁻¹ 62.36 L torr mol ⁻¹ K ⁻¹ 0.08206 L atm mol ⁻¹ K ⁻¹ 1.987 cal mol ⁻¹ K ⁻¹
Planck's constant	h	6.626×10^{-34} J s 4.136×10^{-15} eV s
Planck's reduced constant	$\hbar/2\pi$	1.054×10^{-34} J s 6.582×10^{-16} eV s
Boltzmann constant	k_B	1.38×10^{-23} J K ⁻¹
Stefan-Boltzmann	σ	5.67×10^{-8} W m ⁻² K ⁻⁴
speed of light	c	3.00×10^8 m s ⁻¹
speed of sound (at 20°C)	v_{air}	343 m s ⁻¹
acceleration of gravity	g	9.80 m s ⁻²
gravitational constant	G	6.67×10^{-11} N m ² kg ⁻²
Avogadro's number	N_A	6.022×10^{23} mol ⁻¹
elementary charge	e	1.602×10^{-19} C
Faraday	F	96485 C mol ⁻¹
Coulomb's law constant	k	8.988×10^9 N m ² C ⁻²
Rydberg constant	R	2.178×10^{-18} J

property	symbol	value
electron rest mass	m_e	9.11×10^{-31} kg 0.000549 u 0.511 MeV c ⁻²
proton mass	m_p	1.6726×10^{-27} kg 1.00728 u 938.3 MeV c ⁻²
neutron mass	m_n	1.6749×10^{-27} kg 1.008665 u 939.6 MeV c ⁻²
atomic mass unit	u	1.6605×10^{-27} kg 931.5 MeV c ⁻²
earth mass		5.972×10^{24} kg
earth radius		6.371×10^6 m
moon mass		7.348×10^{22} kg
sun mass		1.989×10^{30} kg
distance earth-moon		3.844×10^8 m
distance earth-sun		1.496×10^{11} m
permittivity of free space	ϵ_0	8.85×10^{-12} F m ⁻¹
permeability of free space	μ_0	$4\pi \times 10^{-7}$ T m A ⁻¹

Some Other Conversion Factors		
1 in	=	2.54 cm
1 lb	=	453.6 g
1 mi	=	5280 ft = 1.609 km
1 gal	=	4 quarts = 231 in ³ = 3.785 L

Some Average Bond Energies (kJ/mol)							
C-H	413	C-C	346	C-Cl	339	C-N	305
O-H	463	C=C	602	C-Br	285	N=N	418
N-H	391	C≡C	835	O=O	498	H-H	436
C-O	358	C=O	799	C≡O	1072	Br-Br	193
H-Cl	432	S-H	347	N≡N	945	Cl-Cl	242
H-Br	366	H-I	299	C≡N	887	I-I	151

Biology Questions (1 – 20)

1. Oxygen is more electronegative than hydrogen, thus H₂O is a
 - A) non-reactive molecule.
 - B) hydrophobic molecule.
 - C) polar molecule.
 - D) linear molecule.
2. RNA is to nucleotide as _____ is to amino acid.
 - A) polysaccharide
 - B) polypeptide
 - C) DNA
 - D) nitrogen
3. Which of the following is not true regarding biological membranes?
 - A) They facilitate communication.
 - B) They are as fluid as salad oil.
 - C) They allow diffusion of CO₂.
 - D) They rely solely on diffusion for passage of materials.
4. A scientist is collecting wild sunflowers across their natural range to study the lipid composition of their seeds. She has discovered that as she travels from the equator northward, there is a graded change in the lipids. Given their different environmental conditions, you would expect plants from the more northern latitudes to make:
 - A) increasing proportions of saturated fatty acids to unsaturated fatty acids.
 - B) increasing proportions of cholesterol to saturated fatty acids.
 - C) increasing proportions of unsaturated fatty acids to saturated fatty acids.
 - D) increasing proportions of both unsaturated fatty acids and saturated fatty acids to cholesterol.
5. A catabolic process _____ the amount of entropy in the universe.
 - A) is dependent on
 - B) increases
 - C) decreases
 - D) maintains
6. A student is frustrated that the chemical reaction they've set up is not going forward. Their lab partner reminds them to add an enzyme, which speeds the reaction considerably. What can be said of this reaction?
 - A) It is not part of a natural, living system.
 - B) It is exergonic.
 - C) The reaction required energy.
 - D) The products contain more energy than the reactants.
7. A key aspect of an electron transport chain is that
 - A) hydrolysis of ATP fuels the chemiosmosis.
 - B) it represents a increase in free energy, as organic fuel gradually donates electrons.
 - C) each electron carrier returns to its reduced form by donating electrons.
 - D) each subsequent electron carrier is more electronegative than the last.
8. In a classic experiment, bacteriophages are first grown in a medium with radioactive sulfur (³⁵S), and then allowed to infect bacteria. After this, the mixture is agitated such that the bacteriophages are shaken loose from the bacteria. The entire mixture is centrifuged such that the heavy bacteria form a pellet in the bottom, and the phages are suspended in the remaining liquid. The pellet and the liquid are then measured for radioactivity. Where should the radioactivity be found?
 - A) In the liquid
 - B) In both the liquid and the pellet
 - C) In the pellet
 - D) In neither the liquid nor the pellet
9. The mRNA for a protein destined to function in the nucleus at maturity will be translated
 - A) at a free ribosome.
 - B) at a ribosome within the nucleus.
 - C) at the rough endoplasmic reticulum.
 - D) at a ribosome bound to the nuclear pore.

10. An operon is
 - A) an efficient way for eukaryotes to regulate which proteins are made.
 - B) a unit of mDNA that a virus inserts into its host.
 - C) a unit of multiple genes all controlled by the same promoter.
 - D) a unit of multiple genes all controlled by the same start codon.
11. A dog that is heterozygous for coat color (Bb) has five puppies. Like their mother, two of the puppies are also heterozygous (Bb) for coat color. Is it likely they inherited both alleles from their mother?
 - A) Yes, the mother passed on her genotype to her puppies, so they are heterozygotes, like her.
 - B) Yes, the law of independent assortment states that the maternal chromosomes will appear together in the same daughter cell, independent of the paternal chromosomes.
 - C) No, the law of segregation states that maternal and paternal alleles separate into different cells during metaphase of meiosis I.
 - D) No, the law of independent assortment states that two alleles of the same gene sort into different chromosomes.
12. Consider a sexually-reproducing animal in which the sperm contain 14 chromosomes. In the cell cycle, just prior to cytokinesis, a cell from this animal's epithelial tissue will contain
 - A) 7 tetrads.
 - B) 14 sister chromatids.
 - C) 28 sister chromatids.
 - D) 56 sister chromatids.
13. The statement that best reflects natural selection is that it depends on
 - A) populations maintaining a certain degree of gene flow with other populations.
 - B) random mutations.
 - C) systematic recruitment of fit individuals into a population.
 - D) mutations occurring as a response to the environment.
14. One of the best lines of evidence that tetrapods evolved from lobed-fin fishes is seen in the discovery of a fossil fish with
 - A) lungs as well as gills.
 - B) scales that function as toes.
 - C) a fin skeleton that includes an elbow and primitive wrist.
 - D) a "walking tail" that evolved into legs.
15. You have sampled some pond water and are examining many single-celled organisms in a powerful microscope. You recognize many as bacteria. What would convince you that you also are seeing members of domain Eukarya?
 - A) Movement due to an outer capsule
 - B) Movement by flagella
 - C) Movement by pseudopodia
 - D) Movement by cilia-like projections
16. Negative pressure within the xylem of a tall tree is greatest during
 - A) warm, humid days.
 - B) cold, foggy days.
 - C) warm, windy days.
 - D) a summer rainstorm.
17. The obligate relationship between a fungus and the ant that grows it for food is considered
 - A) parasitism.
 - B) mutualism.
 - C) herbivory.
 - D) predation.
18. An example of a density-dependent factor affecting population size in mice would be
 - A) infection by mites.
 - B) frequent flooding of the field in which they live.
 - C) Deleterious mutations in their DNA.
 - D) Severity of winter storms.

19. Considering two fruit-eating bird species inhabiting the same area, species 1 will fare best if species 2
- eats the same berries as species 1.
 - is less efficient at finding berries than species 1.
 - eats larger berries than species 1.
 - is more efficient at finding berries than species 1.
20. The burning of fossil fuels releases large quantities of sulfur dioxide and nitrogen oxide into the air. These chemicals mix with water, oxygen, and other chemicals in the atmosphere, and form acidic precipitation. One main reason this is harmful to forests is because
- the acid causes lesions in tree bark.
 - excess sulfur makes leaves more attractive to herbivores.
 - The lower pH disrupts chemiosmosis in the chloroplasts.
 - H^+ is attracted to negatively-charged soil particles.
24. A compound has the formula $C_3H_xO_y$. If the compound is 47.35% carbon by mass, what is the molar mass of this compound?
- 25.4 g/mol
 - 112.0 g/mol
 - 60.1 g/mol
 - 76.1 g/mol
 - 54.0 g/mol
25. A hydrocarbon is burned to produce water and carbon dioxide. If 100.0 g of the hydrocarbon produces 302.9 g of carbon dioxide, what must be the empirical formula for this hydrocarbon?
- C_2H_5
 - CH_2
 - C_3H_8
 - C_2H_6
 - C_3H_7
26. Balance the following reaction with whole numbers as coefficients.
- $$K + KNO_3 \rightarrow K_2O + N_2$$
- What are the respective coefficients?
- $12 : 2 \rightarrow 3 : 1$
 - $10 : 2 \rightarrow 6 : 1$
 - $1 : 1 \rightarrow 3 : 1$
 - $24 : 4 \rightarrow 12 : 2$
 - $8 : 2 \rightarrow 5 : 1$

Chemistry Questions (21 – 40)

21. What is the name for the compound $Mg(CN)_2$?
- magnesium carbonitride
 - manganese nitride
 - manganese cyanide
 - silver carbonitrate
 - magnesium cyanide
22. What is the chemical formula for nitrous acid?
- HNO
 - HNO_3
 - HNO_2
 - H_2NO_3
 - HN_2O_3
23. What is the molar mass of $BaCl_2 \cdot 2H_2O$ to the nearest integer?
- 244 g/mol
 - 218 g/mol
 - 262 g/mol
 - 280 g/mol
 - 237 g/mol
27. Rxn: $Na + KCl \rightarrow NaCl + K$
- What is the maximum amount of NaCl that can be produced when 20.0 grams of Na reacts with 60.0 grams of KCl?
- 80.0 g
 - 47.0 g
 - 50.8 g
 - 54.2 g
 - 58.4g
28. Which of the choices listed has the element with the highest oxidation state ?
- BrO_2^-
 - MnO_2
 - Fe_2O_3
 - NO_3^-
 - CO_3^{2-}

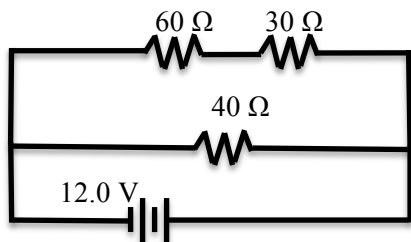
29. AgBr is made via the precipitation reaction
 $\text{AgNO}_3(\text{aq}) + \text{NaBr}(\text{aq}) \rightarrow \text{AgBr}(\text{s}) + \text{NaNO}_3(\text{aq})$
 If 1.85 liters of 0.811 M AgNO₃ are poured into a solution of excess NaBr, and the resulting AgBr precipitate is dried and weighed to be 262 g, what is the percent yield of this reaction?
 A) 100% C) 93% E) 85%
 B) 96% D) 89%
30. What is the electron configuration of arsenic?
 A) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{12} 4p^1$
 B) $1s^2 2s^2 2p^6 3s^2 3p^6$
 C) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9 4p^3$
 D) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8 4p^5$
 E) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^3$
31. Excited electrons in an atom are falling from an excited state of -2.1 eV down to -4.4 eV. What is the wavelength of this emission?
 A) 540 nm C) 470 nm E) 280 nm
 B) 620 nm D) 590 nm
32. What is the molecular geometry for the underlined carbon in acetone, CH₃COCH₃ ?
 A) trigonal pyramidal D) tetrahedral
 B) see-saw E) angular
 C) trigonal planar
33. Molecules X and Y have very similar sizes (radii). X is non-polar and Y is polar. Which of the statements below is true about X and Y?
 A) The geometry of X must be tetrahedral.
 B) X will have a higher solubility in water than that of Y.
 C) Y has no dispersion forces present while X has many dispersion forces.
 D) The dipole moments of X and Y must be identical.
 E) The boiling point of Y is higher than the boiling point for X.
34. A rigid container at 25°C has a gas in it at a pressure of 513 torr. What is the pressure when the temperature is then raised to 75°C ?
 A) 439 torr C) 575 torr E) 1539 torr
 B) 599 torr D) 652 torr
35. Water is added to 75 mL of 6.0 M hydrochloric acid until the total volume is 3.0 L. What is the concentration of this solution?
 A) 0.30 M C) 0.20 M E) 0.10 M
 B) 0.25 M D) 0.15 M
36. What is the pH of a 0.0016 M KOH solution?
 A) 10.75 C) 2.80 E) 1.79
 B) 11.20 D) 12.21
37. Which of the following compounds has the highest normal boiling point?
 A) CH₃CH₂F D) CH₃CH₂CH₃
 B) CH₃OH E) CH₃F
 C) CH₃CH₂OH
38. An (generic) ionic compound is allowed to saturate an aqueous solution via the equilibria
 $\text{MX}_2(\text{s}) \rightleftharpoons \text{M}^{2+}(\text{aq}) + 2 \text{X}^-(\text{aq})$
 What is the value of K_{sp} for MX₂ if the equilibrium concentration of M²⁺ is 0.0035 M ?
 A) 1.2×10^{-5} D) 8.6×10^{-8}
 B) 4.3×10^{-8} E) 1.7×10^{-7}
 C) 4.9×10^{-5}
39. When 9.66 g of sodium is treated with excess oxygen, 87.0 kJ of heat is produced and sodium oxide is formed.
 $4 \text{Na}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2 \text{Na}_2\text{O}(\text{s})$
 What is ΔH_{rxn} for the reaction shown above?
 A) -828 kJ/mol D) -414 kJ/mol
 B) -348 kJ/mol E) -87 kJ/mol
 C) -207 kJ/mol

40. How much heat is required to convert 1.50 moles of water at 28°C into steam at 175°C under 1 atm of pressure ?
- A) 78.7 kJ C) 68.3 kJ E) 54.3 kJ
 B) 12.3 kJ D) 73.3 kJ

Physics Questions (41 – 60)

41. According to Feynman, if only one bit of scientific information could be passed to the next generation, which statement contains the most information in the fewest words?
- A) Energy cannot be created nor destroyed.
 B) Particles behave like waves.
 C) Force equals mass time acceleration.
 D) All things are made of atoms.
 E) Experiment is the sole judge of science.
42. According to Feynman, as water evaporates from a glass of water, which statement is true?
- A) Evaporating molecules transfer momentum to the liquid, heating it.
 B) Evaporating molecules carry away energy from the liquid, cooling it.
 C) Returning molecules divide the energy and cool the liquid.
 D) Returning molecules absorb energy from the liquid and immediately evaporate, cooling the liquid.
 E) The number of evaporating molecules always equals the number of returning molecules.
43. According to Feynman, if the electrical force between two grains of sand were entirely attractive, then the force between the grains separated by 30m would be approximately...
- A) thirty thousand tons.
 B) three million pounds.
 C) thirty million pounds.
 D) thirty million tons.
 E) three million tons.
44. According to Feynman, if an electron were confined in the nucleus of an atom, then the uncertainty principle would require that the electron have a...
- A) very large kinetic energy.
 B) very large potential energy.
 C) very small kinetic energy.
 D) very small potential energy.
 E) none of the above.
45. When a star exhausts its Hydrogen fuel, the inner core of the star collapses, heating the central portion of the star's core and restarting fusion reactions. What element is consumed in this secondary fusion process?
- A) He
 B) Li
 C) C
 D) N
 E) O
46. A plane with an initial velocity of 200m/s accelerates at a constant rate for 3.5s, traversing a distance of 900m during the time it was accelerating. What is the final velocity of the plane?
- A) 110 m/s
 B) 260 m/s
 C) 310 m/s
 D) 510 m/s
 E) 600 m/s
47. A block of mass m_1 sliding to the right at velocity v_1 strikes a mass m_2 which is initially at rest. After the collision, mass m_2 is sliding to the right at velocity v_2 . Which formula below describes the final velocity of mass m_1 ?
- A) $(m_1 v_1 + m_2 v_2)/m_1$
 B) $(m_1 v_1 + m_2 v_2)/m_2$
 C) $(m_1 v_1 - m_2 v_2)/m_1$
 D) $(m_1 v_1 - m_2 v_2)/m_2$
 E) None of the above.

48. Three resistors are connected to a 12.0 V battery as shown. What is the voltage across the 60.0 Ohm resistor in this circuit?



- A) 8.0 V
B) 4.0 V
C) 12.0 V
D) 5.5 V
E) 36.0 V
49. A box of mass 40.0kg is being pushed to the left by a force of 160N across a horizontal floor. The coefficient of friction between the box and the floor is 0.22. What is the acceleration of the box?
- A) 6.2 m/s^2
B) 5.8 m/s^2
C) 4.0 m/s^2
D) 2.2 m/s^2
E) 1.8 m/s^2
50. A 100.0g mass is attached to a spring hanging vertically. The mass is pulled by 15.0cm from the equilibrium point and released from rest. Harmonic motion ensues, and the period of the motion is 1.35s. What is the speed of the mass as it passes through the equilibrium point during each oscillation?
- A) 11.1 cm/s
B) 32.4 cm/s
C) 48.7 cm/s
D) 69.8 cm/s
E) 180 cm/s
51. A Carnot engine has an efficiency of 37%. If the Carnot engine does 60.0J of work each cycle, what amount of heat energy is lost in the exhaust each cycle?
- A) 162 J
B) 102 J
C) 60 J
D) 38 J
E) 22 J
52. A series AC-RLC circuit has a resonant frequency at ω_0 . What is true about the circuit when it is tuned to that resonant frequency?
- A) the circuit is capacitive at resonance
B) the current leads the voltage at resonance
C) the phase angle is zero at resonance
D) the power output is a minimum at resonance
E) none of these are true at resonance
53. Two charges, one positive (Q) and one negative (-2Q), are placed on the x-axis. The negative charge is placed at the origin and the positive charge is placed at $x = r$. At what location on the x-axis is the electric field exactly zero?
- A) $2r/3$
B) $(2 - \sqrt{2})r$
C) $(1 + \sqrt{2})r$
D) $(2 + \sqrt{2})r$
E) $(-2 - \sqrt{2})r$
54. A singly-charged ion beam with a velocity of $2.5 \times 10^5 \text{ m/s}$ passes into a region of uniform magnetic field with a strength of 1200 Gauss. If the diameter of the circle formed by the ion beam is 65.4cm, what is the mass of the ions in atomic mass units (note: $1 \text{ amu} = 1.67 \times 10^{-27} \text{ kg}$).
- A) 63 amu
B) 38 amu
C) 30 amu
D) 25 amu
E) 15 amu

55. A concave mirror with a radius of curvature of 36.0cm is located 13.0cm to the right of a light bulb. If the light bulb is 15.0cm tall, how tall is the image of the light bulb?
- 54.0 cm
 - 39.0 cm
 - 23.5 cm
 - 11.0 cm
 - 8.71 cm
56. A spring with a spring constant of 50.0N/m is compressed 15.0cm by a block with a mass of 0.55kg. Once released, the block slides horizontally across a floor for a distance of 1.2m before coming to rest. What is the coefficient of friction between the floor and the block?
- 0.85
 - 0.58
 - 0.35
 - 0.17
 - 0.087
57. An isotope of Californium-251 decays by the following sequence of radioactive emissions: Alpha, Alpha, Beta-, Gamma, Beta-, Alpha, Gamma, Alpha. What is the isotope produced after this decay chain?
- Protactinium-234
 - Uranium-235
 - Protactinium-235
 - Thorium-232
 - Uranium-236
58. Which fundamental force is responsible for the decay of the neutral pi-meson: $\pi^0 \rightarrow \gamma + \gamma$? The lifetime of the neutral pi-meson is 8.4×10^{-17} s
- Gravitational Force
 - Weak Force
 - Electromagnetic Force
 - Strong Force
 - Higgs Force
59. A 1000 W radio antenna produces 95.7MHz radio waves that expand uniformly in all directions. What is the maximum strength of the electric field of the radio waves at a location 400.0m from the antenna? (note: $\epsilon_0 = 8.854 \times 10^{-12} \text{ Nm}^2/\text{C}^2$)
- 0.053 N/C
 - 0.38 N/C
 - 0.43 N/C
 - 0.61 N/C
 - 1.2 N/C
60. A certain atom has four energy states: a ground state at 0.0eV, and three excited states at energies of 1.77eV, 3.42eV, and 6.14eV. Which of the following photon wavelengths would you not expect to detect emitted from a collection of these atoms? (note: $h = 6.626 \times 10^{-34} \text{ Js}$)
- 202 nm
 - 284 nm
 - 456 nm
 - 514 nm
 - 752 nm